VINUNIVERSITY

**Test duration: 1 hour and 30 minutes**

**Name:** **Student ID:**

# Midterm Practice Test 5 – Solutions – Fall 2026

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**COMP3040/5030 Computer Vision** Unofficial practice test, prepared by Lê Tăng Phú Quý

## Instructions:

- Academic honesty is required in all work you submit to be graded. You should **NOT** copy or share your solutions with other students to avoid plagiarism issues.
- For short answer questions, please make your answers as **concise** as possible.
- This is a closed-book exam and NO Internet access is allowed. You are allowed 1 page of **hand-written** note (both sides).
- Drafting paper and pen/pencil are allowed. We will provide the drafting paper to you.
- Late submission without an approved extension will not be graded at all.
- You must **RETURN** this hard copy of the midterm exam at the end of the exam period.
- During the exam, you are **NOT** allowed to leave the room or/and use the restroom.
- You are required to **NOT** use your mobile phone in the exam room. Please put it on the table or in your bag.
- Good luck!

**About the exam**: There are **05** problems with **100 points** total, and some have multiple parts. Please check through the pages before you start.

| Problem | Content | Point (Undergraduates) | Type |
|---|---|---|---|
| 1 | Fundamentals of Computer Vision | 20 | short answer |
| 2 | Image Filtering | 25 | long answer |
| 3 | Feature Detection and Matching | 25 | long answer |
| 4 | Fitting and Image Transformations | 25 | long answer |
| 5 | System Design | 05 | long answer |
| **Total** | | **100** | |

## Problem 1. Fundamentals of Computer Vision

Give a short answer for each of the following question:

**1.1.** A 9 Hz wave is sampled at 12 Hz. Which frequency do the samples appear to have? What is the minimum alias-free sampling rate? (5 pts)

*Solution.* Folding modulo $f_s$: $|9 - 12| = 3$ Hz. One needs $f_s > 2 \cdot 9 = 18$ Hz.

**1.2.** When should you prefer **RANSAC**, and when the **Hough transform**? Give one distinctive advantage of each. (5 pts)

*Solution.* RANSAC: many-parameter models (homography 8, F-matrix), one dominant model, moderate outliers — memory does not grow with the parameter count. Hough: few parameters (lines, circles), **multiple instances** at once, broken lines — deterministic, one peak per instance.

**1.3.** Write the reflection across the horizontal ($x$) axis and its determinant. What does a negative determinant mean? Composing the reflections across both axes gives which transformation? (5 pts)

*Solution.* $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$, $\det = -1$: area preserved but **orientation reversed** (clockwise $\leftrightarrow$ counterclockwise). $\mathrm{diag}(-1, 1) \cdot \mathrm{diag}(1, -1) = -I$: the flip through the origin = a **180°** **rotation** ($\det = +1$).

**1.4.** Two images of the same scene and viewpoint, but the second is brighter and has higher contrast. Which distance should compare them? Explain. (5 pts)

*Solution.* ZNCC (zero-mean normalized cross-correlation) — invariant to $aI + b$; or compare normalized gradients/descriptors. SSD/L2 would report a large difference purely from lighting.

## Problem 2. Image Filtering

**2.1.** A $7 \times 7$ image has a 2-pixel-thick **horizontal** line in rows 3–4 (value 1, else 0). Convolve **vertically** with $[-1, 0, 1]^\top$ (zero padding). (a) Write one column of the result. (b) How many local maxima (along columns)? How many for |response|? (c) How does the result change with correlation instead of convolution? (13 pts)

*Solution.* (a) Convolution flips the kernel: $o[i] = f[i-1] - f[i+1] \Rightarrow [0, 0, -1, -1, 1, 1, 0]^\top$. (b) One positive plateau per column (rows 4–5) $\Rightarrow$ **7** maxima; for $|o|$: the top and bottom edges $\Rightarrow$ **14**. (c) Correlation: $o[i] = f[i+1] - f[i-1] \Rightarrow$ every sign flips, $[0, 0, 1, 1, -1, -1, 0]^\top$: the maxima move to the top edge; $|o|$ is unchanged.

**2.2.** (a) By the $2\lceil 3\sigma \rceil + 1$ rule, what kernel size does $\sigma = 1.4$ need? Why renormalize the sum to 1 after truncation? (b) Blurring a uniform white image (255) with a Gaussian and zero padding: what happens at the border? Which border mode fixes it? (12 pts)

*Solution.* (a) $\lceil 4.2 \rceil = 5 \Rightarrow 11 \times 11$. Samples cut outside $\pm 3\sigma$ (and discretization) make the sum $\neq 1 \Rightarrow$ without renormalizing the image darkens/brightens. (b) Zero padding mixes in zeros $\Rightarrow$ a **dark band** about $3\sigma$ wide along the border. Replicate/reflect (OpenCV's default

`BORDER_REFLECT_101`) keep exactly 255.

## Problem 3. Feature Detection and Matching

**3.1.** After NMS, an 8-connected chain of pixels along a curve has (normalized) magnitudes $[0.9, 0.5, 0.45, 0.5, 0.2, 0.6, 0.95]$. Canny thresholds: low $= 0.4$, high $= 0.8$. (a) Classify each pixel. (b) Which pixels are edges after hysteresis? (c) What would a single threshold of $0.8$ give? Of $0.4$ over the whole image? (12 pts)

*Solution.* (a) Strong: indices 0, 6; weak: 1, 2, 3, 5; dropped: 4 (0.2). (b) 1–3 connect to 0 (strong), 5 connects to 6 $\Rightarrow$ edges: **0, 1, 2, 3, 5, 6**; the chain breaks at 4. (c) Threshold 0.8: only 2 pixels — fragmented edges. Threshold 0.4: keeps this chain but also every isolated weak noise segment elsewhere. Hysteresis seeds with the high threshold and extends with the low one.

**3.2.** SIFT keypoint localization (1-D in $x$ for simplicity): $D(-1) = 0.2$, $D(0) = 0.5$, $D(1) = 0.4$ (image in $[0, 1]$). (a) With central differences, find $\hat{x}$ and $D(\hat{x})$. Keep or reject with the 0.03 threshold? (b) Spatial Hessian $\begin{bmatrix} 4 & 1 \\ 1 & 3 \end{bmatrix}$: does it pass the edge test with $r = 10$? (13 pts)

*Solution.* (a) $D' = (0.4 - 0.2)/2 = 0.1$, $D'' = 0.4 + 0.2 - 1.0 = -0.4 \Rightarrow \hat{x} = -D'/D'' = 0.25$ (< 0.5, no resampling); $D(\hat{x}) = D + \frac{1}{2}D'\hat{x} = 0.5125 > 0.03 \Rightarrow$ **keep**. (b) $\mathrm{tr}^2 / \det = 49/11 \approx 4.45 < 12.1$, $\det > 0 \Rightarrow$ **passes** (blob-like).

## Problem 4. Fitting and Image Transformations

**4.1.** Homography estimation with the DLT. (a) Write the two rows of $A$ for the pair $(2, 3) \to (5, 7)$. (b) Why are at least 4 pairs, no three collinear, required? (c) Why normalize coordinates (Hartley) before solving with the SVD? (12 pts)

*Solution.* (a) $[-2, -3, -1, 0, 0, 0, 10, 15, 5]$ and $[0, 0, 0, -2, -3, -1, 14, 21, 7]$. (b) 8 DOF, 2 equations per pair $\Rightarrow$ 4 pairs; three collinear points give dependent equations $\Rightarrow A$ is rank-deficient, no unique solution. (c) Pixel coordinates ~$10^3$ put columns like $ux$ ~$10^6$ next to columns of 1 $\Rightarrow A$ is badly conditioned; normalizing (centroid 0, mean distance $\sqrt{2}$) balances the columns, then denormalize $H = T_d^{-1}\tilde{H}T_s$.

**4.2.** Vector $\mathbf{x} = (5, -2, 4)$. A new coordinate system is obtained from the original by translating by $(1, 2, 3)$, followed by a rotation about the $Z$ axis by $-90°$. What are $\mathbf{x}$'s coordinates in the new system? Write the $4 \times 4$ homogeneous matrix performing this change of coordinates. (13 pts)

*Solution.* $\mathbf{x} - \mathbf{t} = (4, -4, 1)$. $R_z(-90°) = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$; new coordinates $= R^\top(\mathbf{x} - \mathbf{t}) = (0 \cdot 4 - 1 \cdot (-4),\ 4,\ 1) = (4, 4, 1)$. Matrix: $\begin{bmatrix} R^\top & -R^\top\mathbf{t} \\ \mathbf{0}^\top & 1 \end{bmatrix}$ with $R^\top\mathbf{t} = (-2, 1, 3) \Rightarrow$ last column $(2, -1, -3)$. (The "move the point" reading gives $R(\mathbf{x}+\mathbf{t}) = (0, -6, 7)$ — state your assumption.)

## Problem 5. System Design

**5.** A fixed traffic camera looks obliquely down a straight road. Design a system that **estimates each vehicle's speed** (km/h) from video alone, given only the real-world size of a few road markings. (5 pts)

*Solution.* (1) One-off calibration: $\geq$4 marking points with known road-plane coordinates (m) $\Rightarrow$ an image $\rightarrow$ ground **homography** (DLT/RANSAC) since the road is planar. (2) Per frame: background subtraction (temporal mean/median) or a vehicle detector $\rightarrow$ boxes; take the ground-contact point (bottom-centre of the box). (3) Track vehicles across frames (feature/ZNCC matching or optical flow, nearest assignment). (4) Map contact points through $H$ to metres, fit a line (least squares/RANSAC) of distance vs time $\Rightarrow$ speed = slope $\times$ fps $\times$ 3.6. Caveat: only points **on the road plane** obey the homography (a car roof would be wrong because of its height).